



Pimpri Chinchwad Education Trust (PCET)
Pimpri Chinchwad College of Engineering

An ISO 9001:2015 Certified Institute, NBA Accredited, Accredited by NAAC with 'A' grade

Department of Computer Engineering Regional

Major Project

Presentation on

**Identification of Medicinal Plants using Image
Processing with Deep Learning**

Presented By

Piyush Hole (122B2D068)

Atharv Nikam (121B1D066)

Sufiya Sayyad (122B2D070)

Harshal Naik (121B1D062)

Guided By

Prof. Rohini. Sarode

Problem Statement

The accurate identification and authentication of medicinal plants are critical for healthcare, conservation, and traditional medicine practices to develop and implement a robust and efficient system for the automated detection and classification of medicinal plant species using image processing and machine learning techniques.

There are several common problems related to medical plant identification today:

- 1) Species confusion
- 2) Limited knowledge

Motivation

- 1) **Inaccurate Plant Identification:** The proposed system aims to provide accurate plant identification, reducing the risk of misidentification.
- 2) **User-Friendly Interface:** **Voice technology** and **Regional language** support make the system more accessible to a wider range of users, including those with limited literacy or technical skills.
- 3) Such a system can have far-reaching positive effects, from conserving biodiversity to **preserving traditional knowledge** and **improving healthcare access**.

Objectives

- 1) To build the **capability of identifying medicinal plants accurately** in the existing plant identification system.
- 2) To **improve the performance** of existing plant identification systems by providing the accurate data to training model with various different algorithms.
- 3) To have **robust** plant identification systems that can tackle a variety of issues by adding technologies such as voice technology and regional support with plant description.
- 4) To build a **computationally lighter** plant identification model.

Literature Review

Deep Learning Based Methods

Sr No.	Title	Year	Methodology	Strengths	Weakness
1	A Novel Computer Vision Model for Medicinal Plant Identification Using Log-Gabor Filters and Deep Learning Algorithms	2022	The proposed CNN model (OTAMNet) created is based on the DenseNet architecture since it has the following advantages: since error signals can be easily transferred to older levels more directly, there is a significant gradient flow; more diversified features can be extracted because each layer in DenseNet receives all preceding layers as input as opposed to standard CNN models where the classifier uses the most complex features.	Highest accuracy achieved among the papers considering hand crafted features techniques	Method is assessed only on leaf dataset.
2	An AI Based Approach for Medicinal Plant Identification Using Deep CNN Based on Global Average Pooling	2022	The proposed CNN model has been able to correctly classify the images of the medicinal plants with overall accuracy rates of 99.66, 99.32, and 99.45% when the input images are 64×64 , 128×128 , and 256×256 pixels	Inter-database evaluation is done.	Only deep learning CNN approach is done

Literature Review

Machine Learning Based Methods

Sr No.	Title	Year	Methodology	Strengths	Weakness
3	DeepHerb: A Vision Based System for Medicinal Plants Using Xception Features	2021	The research focuses on automated identification of medicinal herbs using transfer learning technique in deep learning. Due to the unavailability of the herb dataset, efforts taken to construct a custom medicinal leaf dataset	The features are classified using two machine learning techniques such as ANN and SVM.	Not given the respective plant description to the label
4	Medicinal Plant Identification using Image Processing Technique	2020	Implemented a technique for medicinal plant identification using random forest algorithm, an ensemble supervise machine learning algorithm based on color ,texture and geometrical features to identify the correct species of medicinal plant. the combination of shape, color and texture features result in correct leaf identification accuracy of 94.54 %.	Proposed method is simple and efficient in terms of computation.	Feature length is large.

Literature Review

Deep Learning Based Methods

Sr No.	Title	Year	Methodology	Strengths	Weakness
5	MedicPlant: A mobile application for the recognition of medicinal plants from the Republic of Mauritius using deep learning in real-time	2021	The first step is to create a deep learning model. Thus, starting with the Inception-v3 pre-trained model developed at Google, we added a new layer which can recognise medicinal plants. The Inceptionv3 model is a 42-layer deep learning network that has been trained on a subset of the ImageSet dataset consisting of 1000 classes with 1000 images in each category. This method of training is known as transfer learning.	Use of Inception-v3 pre-trained model with the technique of transfer learning	Work only for mauritious plant names.

Sr No	Title	Year	Methodology	Strengths	Weakness
6	A method for accurately segmenting images of medicinal plant leaves with complex backgrounds	2018	The paper proposes a relatively accurate method of segmenting the leaf images of medicinal plants with complex backgrounds. The method directly takes photos of leaves on branches for the purpose of machine identification and protects the plant from leaf picking; but it also encounters higher difficulties in segmentation as branches, soil and other leaves appear inevitably in the background.	Method of Segmentation	Only segmentation method is applied.
7	Ayurvedic Plant Species Detection System Using Image Processing	2022	The proposed system is vision based, uses image processing techniques and inputs from a computer or laptop camera. Our system is specific towards detecting different species of the same plant.	Use of a single convolutional network makes up the SSD architecture, which learns to predict bounding box positions and classify these places in a single run.	Method is explored on only one dataset.

Flowchart of Proposed Method

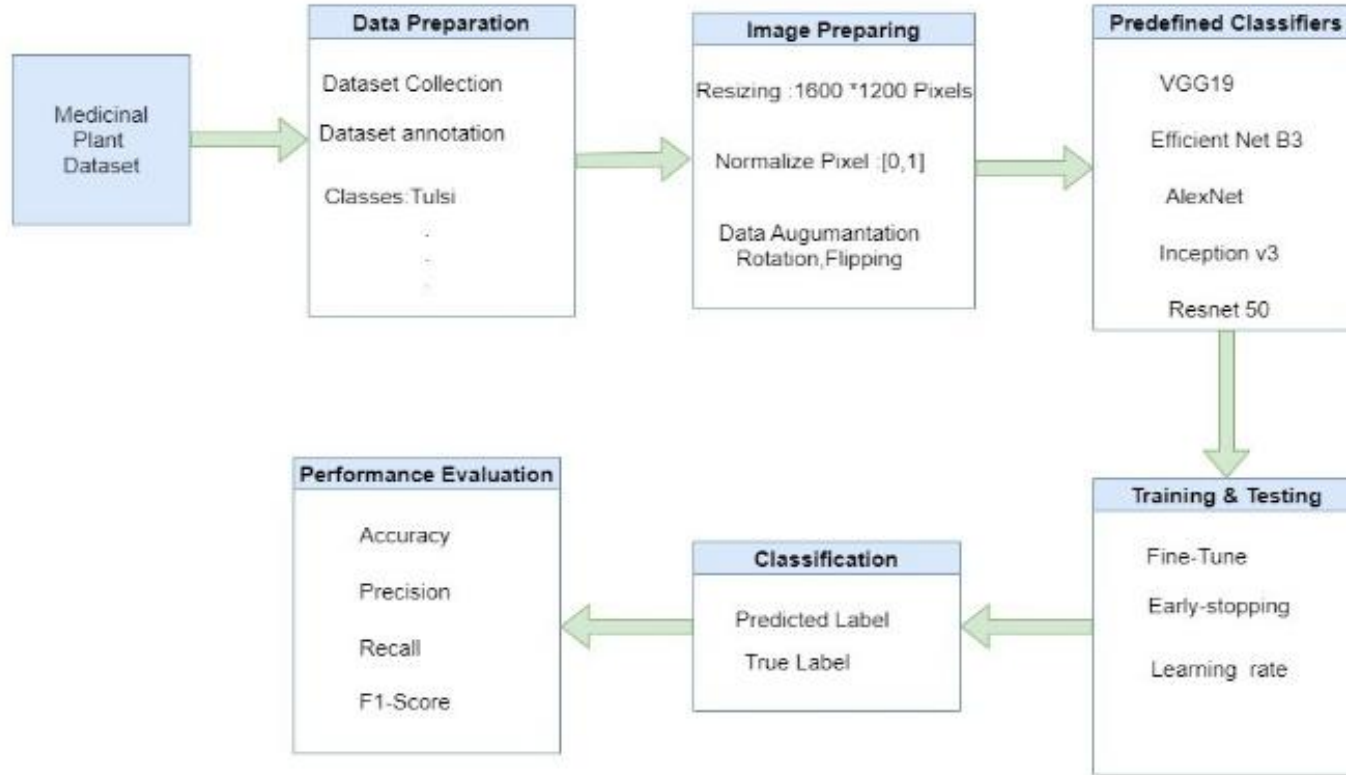


Fig.1: Flowchart

Algorithm Steps

Step 1: **Data Collection & Preparation** – Gather and preprocess medicinal plant images.

Step 2: **Feature Extraction** – Use a fine-tuned VGG19 model to extract key features.

Step 3: **Deep Learning Model** – Fine-tune pre-trained models like VGG for better accuracy.

Step 4: **Model Training** – Apply transfer learning and validate using test datasets.

Step 5: **Testing & Validation** – Ensure accuracy in image recognition, voice commands, and language support.

Step 6: **Final Classification** – Identify plant species based on trained model predictions.

Review 1 Suggestions

- Dataset Improvement:
 - - Increase dataset size with more diverse images
 - - Ensure high-quality, well-labeled images
- Real-Time Plant Detection:
 - - Implement real-time plant identification using a camera feed
- Report Generation:
 - - Develop an automated report generation system
 - - Include plant details, medicinal uses, and plant information
 - - Provide export options (PDF) for user convenience

Improvements

- **Enhanced Dataset:**
 - Collected additional high-quality images of various background conditions
 - Performed data augmentation for better model generalization
 - Improved labeling and balanced the dataset
- **Real-Time Plant Detection:**
 - Integrated live camera-based plant recognition
 - Optimized model inference for real-time performance
 - Tested on different lighting and background conditions
- **Automated Report Generation:**
 - Implemented report generation with plant details and confidence scores
 - Added export options (PDF) for easy sharing
 - Improved UI for better user experience

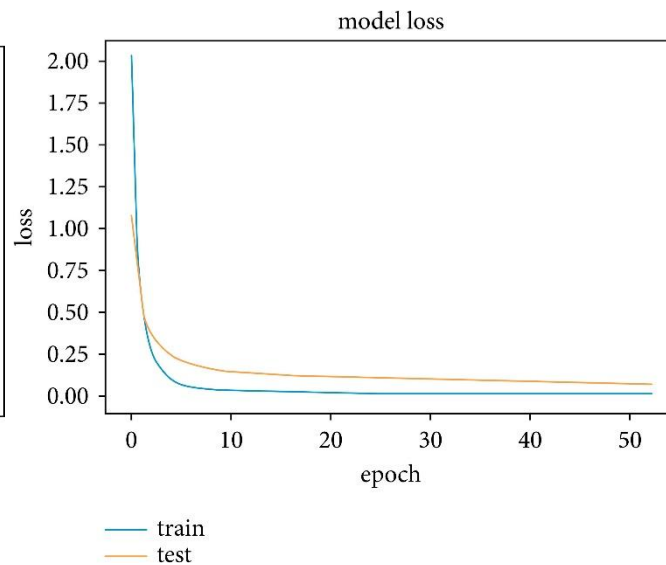
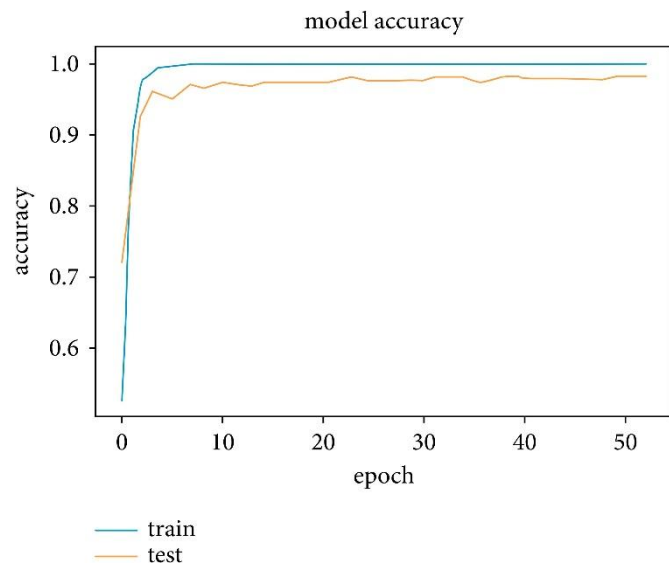
Results Obtained:

Model	Accuracy (%)	Avg Precision (%)	Avg Recall (%)	Avg F1-Score (%)
VGG19	99.24	98.06	99.12	98.89
EfficientNetB3	98.75	97.54	98.02	98.25
ResNet50	92.4	91.45	90.14	92
Alex-Net	90.65	89.2	90.14	90.95
Inception-V3	85	84.22	86.12	85.44

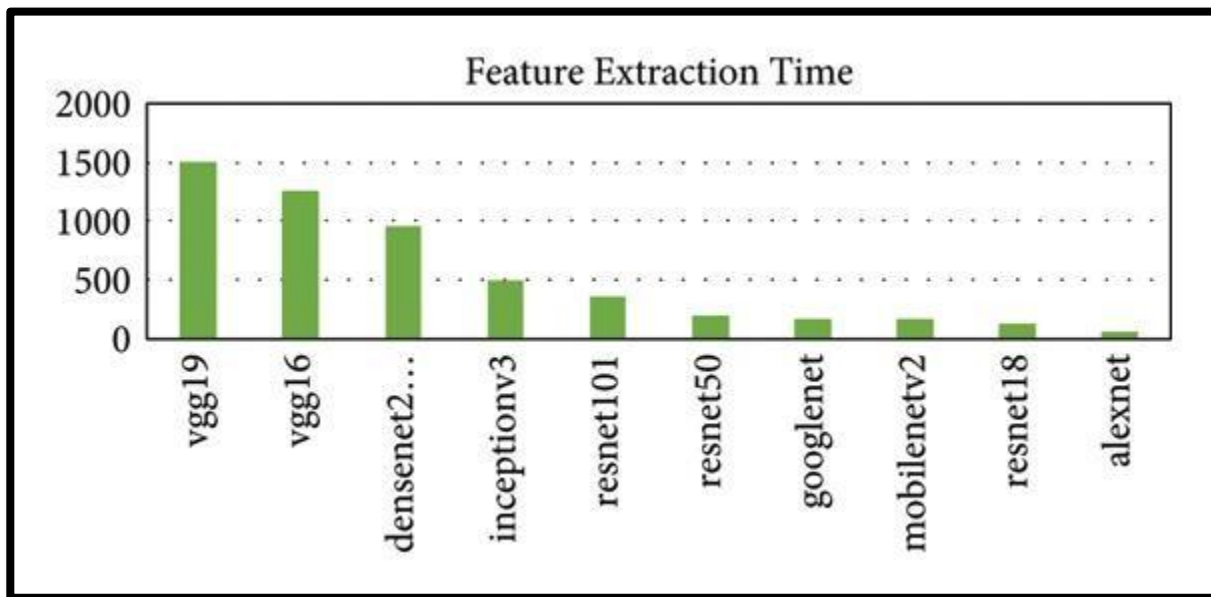
- VGG19 emerged as the best model for our medicinal plant leaf identification project, achieving **99.24% accuracy, 98.06% precision, 99.12% recall, and an F1-score of 98.89%**.

VGG19, a deep convolutional neural network with 19 layers, is known for its simple yet highly effective architecture. Its uniform 3×3 convolutional layers allow it to learn intricate patterns in images while maintaining computational efficiency. The model performed exceptionally well in our use case, accurately distinguishing between different medicinal plant leaves. Given its high accuracy and balanced precision-recall scores, VGG19 was chosen as the final model for deployment.

Results Obtained :

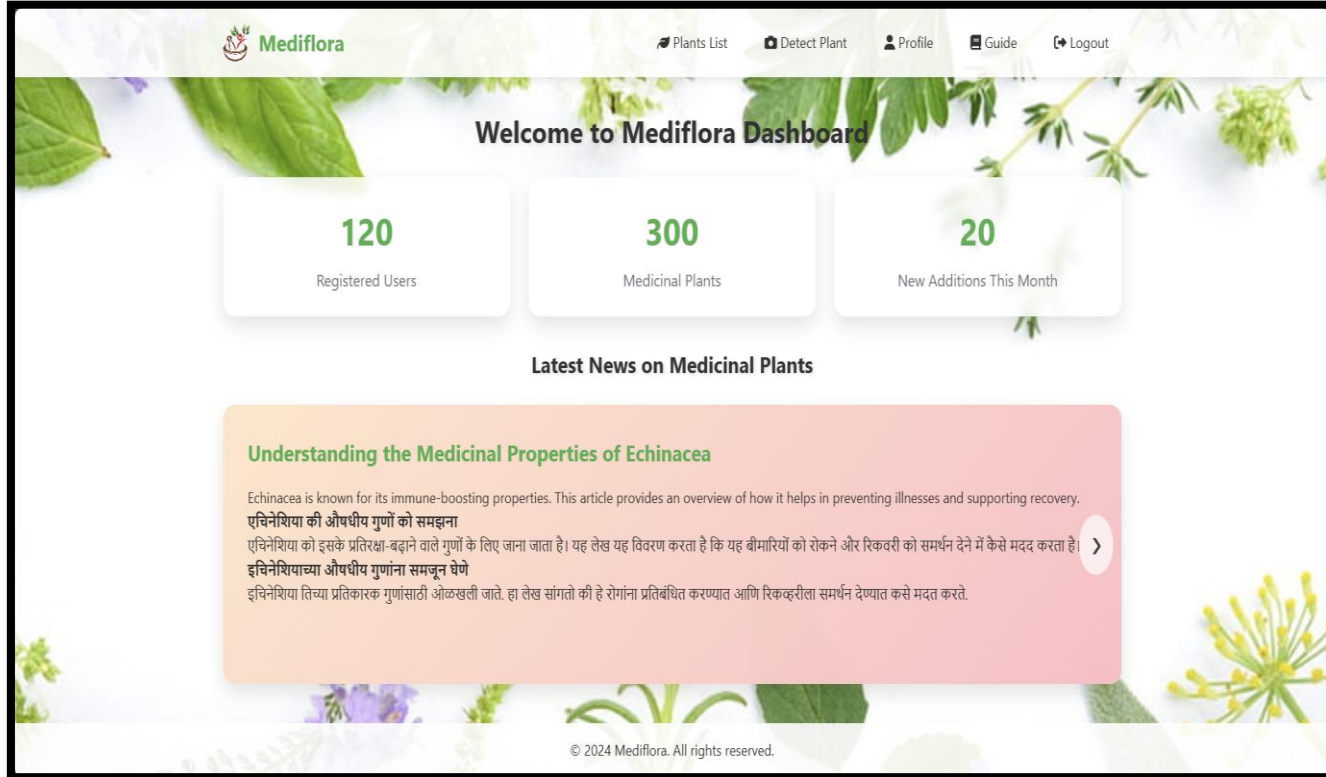


Results Obtained:



Result

Mediflora Dashboard



Plants Details

Plant Categories

🌸 Flowering Plants

🌳 Trees

🌿 Shrubs

🌱 Herbs

🌊 Aquatic Plants

🏠 Go to Dashboard

Back to Plants

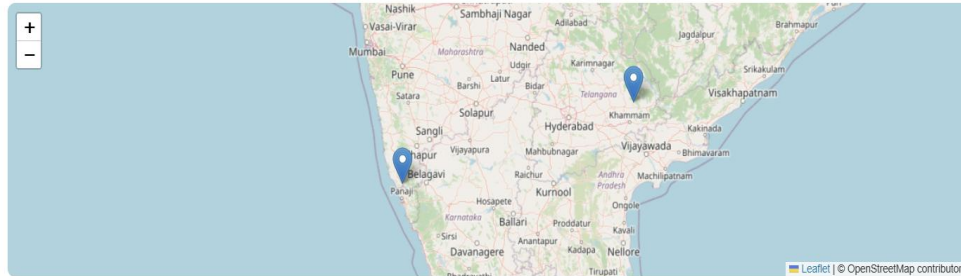
Basale



Basale has an anti-inflammatory activity and wound healing ability. It can be helpful as a first aid, and the leaves of this plant can be crushed and applied to burns, scalds, and wounds to help in healing of the wounds.

Regions: Western Ghats, Eastern Ghats, Eastern Himalaya

Location: Longitude 73.8372, Latitude 15.815



[View on Wikipedia](#)

Plant Recognition in Multiple Languages

Plant Recognition and Description

Upload or Capture Plant Image

Capture Image

Choose File SA-S-015.jpg

Predict Image

Reset

Choose Language & Get Description

Choose Language:

English

Speak Plant Name

Predicted Plant and Description:

Sandalwood

Sandalwood: Sandalwood is used for treating the common cold, cough, bronchitis, fever, and sore mouth and throat. It is also used to treat urinary tract infections (UTIs), liver disease, gallbladder problems, heatstroke, gonorrhea, headache, and conditions of the heart and blood vessels (cardiovascular disease).



Plant Recognition and Description

Upload or Capture Plant Image

Capture Image

Choose File SA-S-015.jpg

Predict Image

Reset

Choose Language & Get Description

Choose Language:

Hindi

Speak Plant Name

Predicted Plant and Description:

Sandalwood

चंदन: चंदन का उपयोग सामान्य सर्दी, खांसी, ब्रोंकाइटिस, बुखार और मुँह और गले में दर्द के इलाज के लिए किया जाता है। इसका उपयोग मूत्र पथ संक्रमण (UTIs), यकृत रोग, पित्ताशय की समस्याएँ, गर्मी से संबंधित बीमारियाँ, गोनोरिया, सिरदर्द और हृदय और रक्त वाहिनियों की समस्याओं (कार्डियोवैस्कुलर रोग) के इलाज के लिए भी किया जाता है।



Project Feasibility

❑ **Technical Feasibility**

Utilization of deep learning models (CNN, VGG-19) for image-based plant identification.
Integration of voice recognition and regional language support for user accessibility.

❑ **Economic Feasibility**

Cost-effective implementation using open-source frameworks (Python, TensorFlow).
Potential for commercialization in healthcare, agriculture, and education sectors.

❑ **Operational Feasibility**

User-friendly interface allowing easy plant identification through image and voice inputs.
Scalability to support more plant species and multiple languages over time.
Regular dataset updates to improve identification accuracy.

Project Scope

❑ **Core Functionality**

Medicinal plant identification using image processing and deep learning.

Feature extraction and classification using CNN models.

Voice-based interaction for accessibility in multiple regional languages.

❑ **Target Audience**

Farmers, herbal medicine practitioners, researchers, and educational institutions.

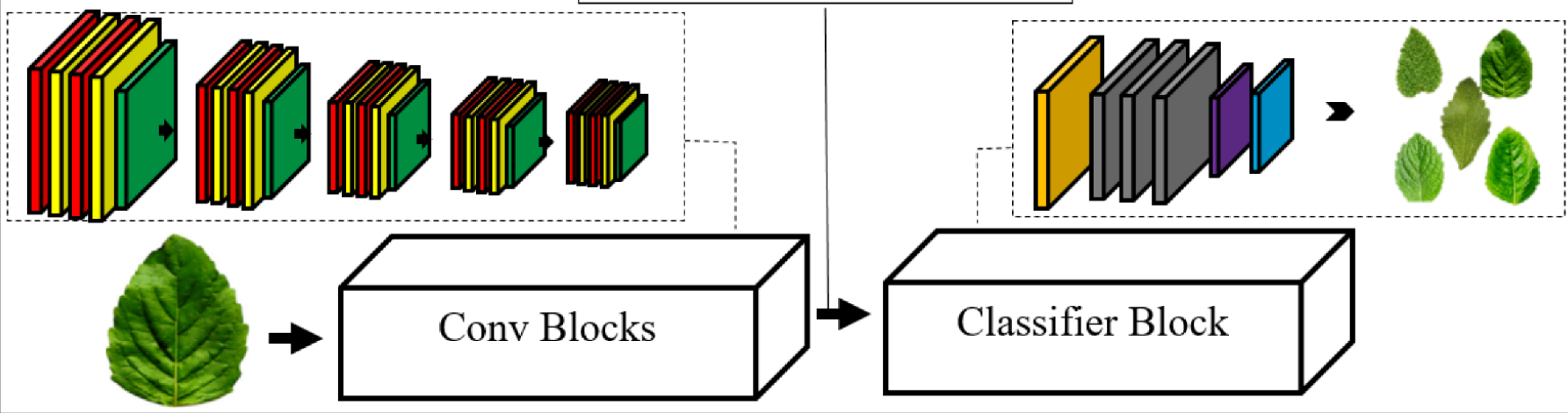
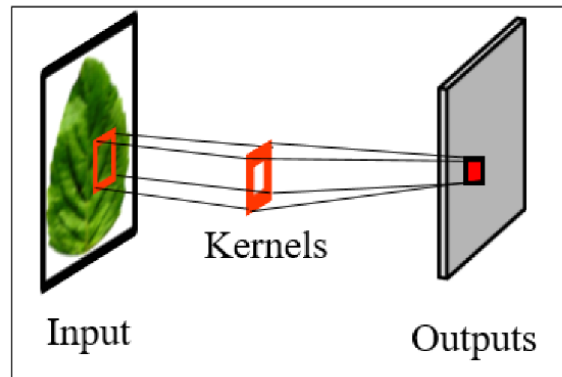
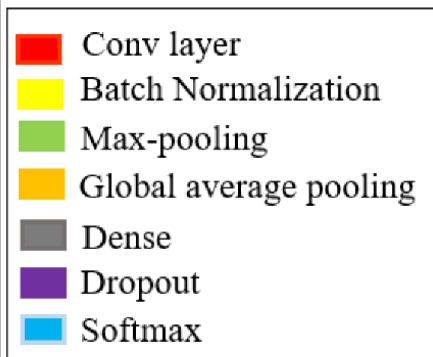
General users interested in identifying medicinal plants accurately.

❑ **Expansion Possibilities**

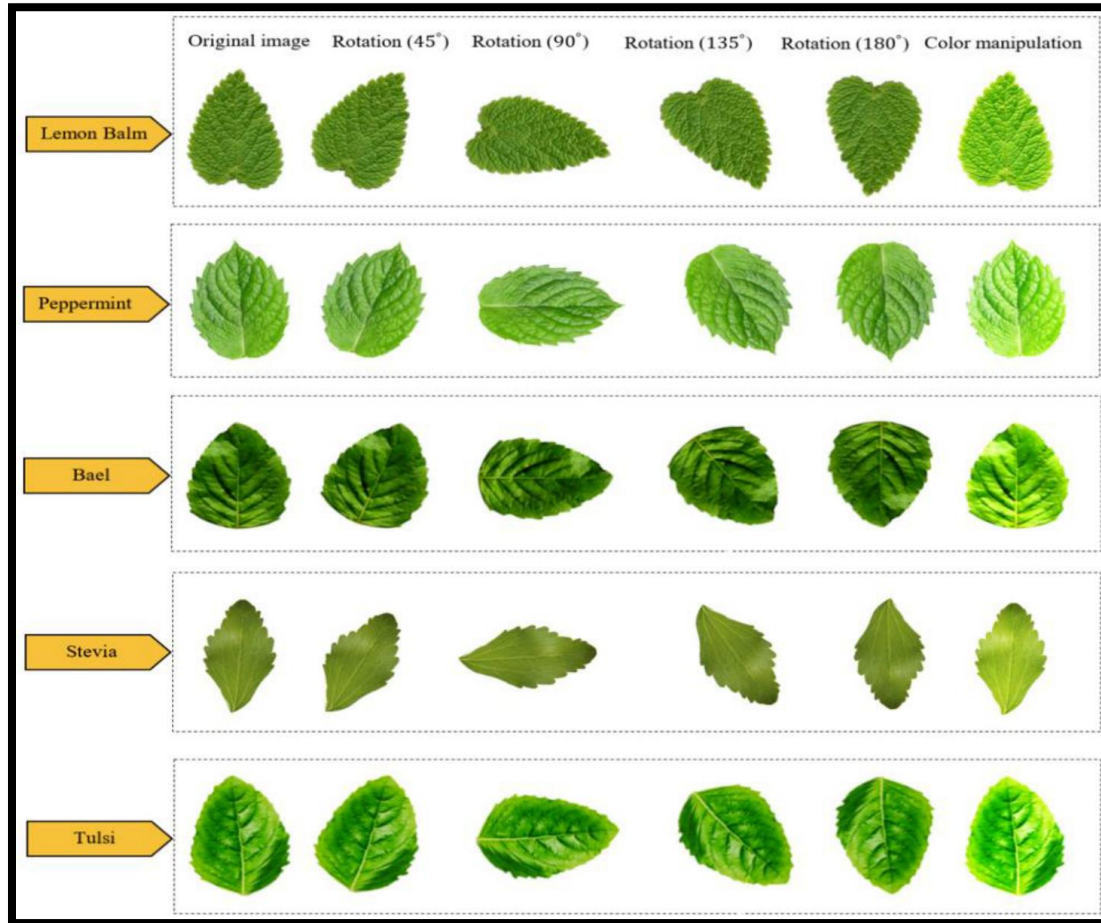
Integration of an AI chatbot for plant care guidance.

Database expansion to include global medicinal plant species.

Architecture of CNN model



Data Augmentation Technique :



Projects Requirement

1. Hardware Requirements

- **CPU:** Intel i5/i7 or AMD equivalent
- **GPU:** NVIDIA GPU (RTX 2060 or higher recommended) for faster training
- **RAM:** Minimum 8GB (16GB recommended)
- **Storage:** At least 50GB free space for dataset and model storage
- **Camera:** High-resolution camera for real-time detection

2. Software Requirements

- **Operating System:** Windows / Linux / macOS
- **Programming Language:** Python
- **Deep Learning Framework:** TensorFlow / Keras / PyTorch
- **Libraries:** OpenCV, NumPy, Pandas, Matplotlib
- **Database :** MongoDB

3. Dataset Requirements

- High-resolution images of medicinal plant leaves
- Properly labeled dataset for training and testing
- Data augmentation techniques to improve model generalization

Conclusion

The project "Identification of Medicinal Plants Using Image Processing with Voice Technology" represents a groundbreaking endeavour at the intersection of plant identification, herbal medicine, and technological innovation. Through the fusion of image processing, deep learning models (including VGG-19), and voice technology, the project offers the potential for a transformative shift in how we identify, utilize, and conserve medicinal plants. Key achievements include the development of an efficient system for precise identification, reducing dependence on expert input and enhancing accessibility. Voice technology enriches user experience, enabling natural language interactions and expanding educational opportunities. This project's promise extends to academia and public awareness campaigns, fostering greater understanding of medicinal plants.

References

[1] Stephen Opoku Oppong,¹Frimpong Twum,²James Ben Hayfron-Acquah,²and Yaw Marfo Missah²

¹Department of ICT Education, University of Education, Winneba, Ghana September-2022

[2] Rahim Azadnia 1 , Mohammed Maitham Al-Amidi 2 , Hamed Mohammadi 3 , Mehmet Akif Cifci 4 , Avat Daryab 5 and Eugenio Cavallo November-2022.

[3] S. ROOPASHREE 1,2, (Member, IEEE), AND J. ANITHA³ ¹Department of Computer Science and Engineering (AI&ML) and (Cybersecurity), Faculty of Engineering and Technology, JAIN (Deemed-to-be University), Bengaluru 560041, India October-2021

[4] Nayana G. Gavhale¹, Dr. A. P. Thakare², ² Department of Electronics and Telecommunication , Sipna College of Engineering and Technology Amravati, Maharashtra May-2020.

[5] Sameerchand Pudaruth¹ , Mohamad Fawzi Mahomoodally² , Noushreen Kisson³ , Fadil Chady⁴ ^{1,4}ICT Department, University of Mauritius, Mauritius ^{2,3}Department of Health Sciences, Faculty of Medicine and Health Sciences, University of Mauritius, Mauritius 2021.

[6] Liwen Gao ^a , Xiaohua Lin ^b School of Medical Information Engineering, Guangzhou University of Chinese Medicine, Guangzhou, China December 2018.

[7] Anupriya Varshney, Kunal Dongare, Muskan Shrivastava, Janhavi Naik, Dr. Sanjesh Pawale, 2022.

THANK YOU !!!!